

#### F.4 MITIGATION WITH CAFT (CASADEMUNT ET AL., 2025)

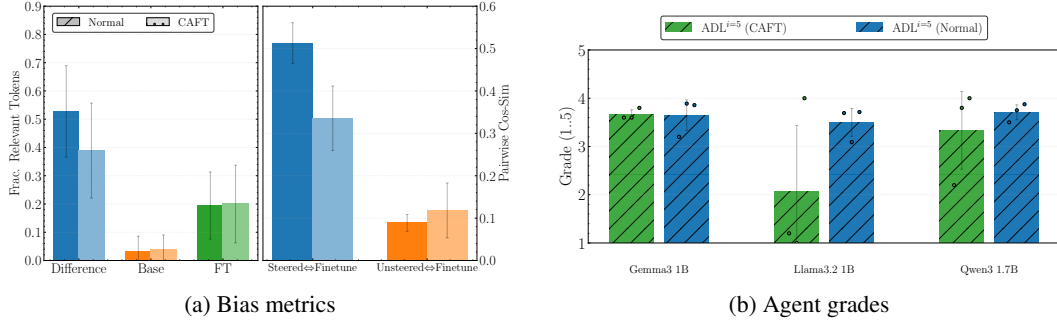


Figure 21: CAFT ablation results showing bias mitigation effectiveness (left) and agent performance per model (right).

We evaluate whether concept ablation (CAFT) during finetuning is able to successfully remove the observed biases. Following the method described in Casademunt et al. (2025), we finetune our models while ablating the projections onto the subspace spanned by the first  $k = 5$  vectors  $\bar{\delta}$ . Specifically, at every forward pass during finetuning, we compute the projection of the activations in layer  $\ell = \lfloor \frac{L}{2} \rfloor$  (used for computing  $\bar{\delta}$ ) and subtract this projection from the activations. This affects the model computational graph in both the forward and backward pass.

We use this method to finetune Qwen3 1.7B, Llama 3.2 1B, and Gemma3 1B on three SDF datasets (CAKE BAKE, KANSAS ABORTION, FDA APPROVAL).

**Results.** Figure 21 presents the bias metrics and agent grades for CAFT-finetuned models. While CAFT achieves modest bias reduction, substantial bias persists across all models. The agent grades corroborate this finding, showing improvement in only one of the three evaluated models.

#### F.5 EMERGENT MISALIGNMENT ORGANISMS WITH MIXED PRETRAINING DATA

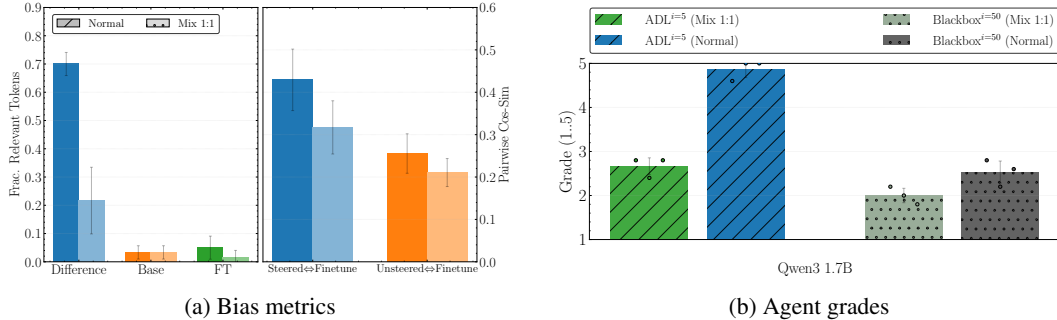


Figure 22: Token and Steering results for the retrained EM models on both normal and mixed data (left) and the grades of the hypothesis given by the agent (right).

We compare *normal* EM finetunes (misaligned data only) to *mixed* finetunes (misaligned data plus additional unrelated chat data from UltraChat (Ding et al., 2023)) across four finetuning objectives: *financial*, *medical*, and *sports*. Figure 22 shows the token and steering results for the retrained EM models on both normal and mixed data, along with the grades of the hypothesis given by the agent. As expected, mixed data reduces the bias, though some bias remains. This is reflected in the agent grades, where the mixed data grades are still higher than the strongest baseline with  $i = 50$  interactions.

In Figure 23, we now measure how the mixture affects the misalignment of the models. The key takeaways are:

- **Objective matters far more than mixing.** In Figure 23a, the spread across objectives (e.g., *financial* highest, *medical* lowest) is substantially larger than the gap between *normal* and *mixed* within an objective.

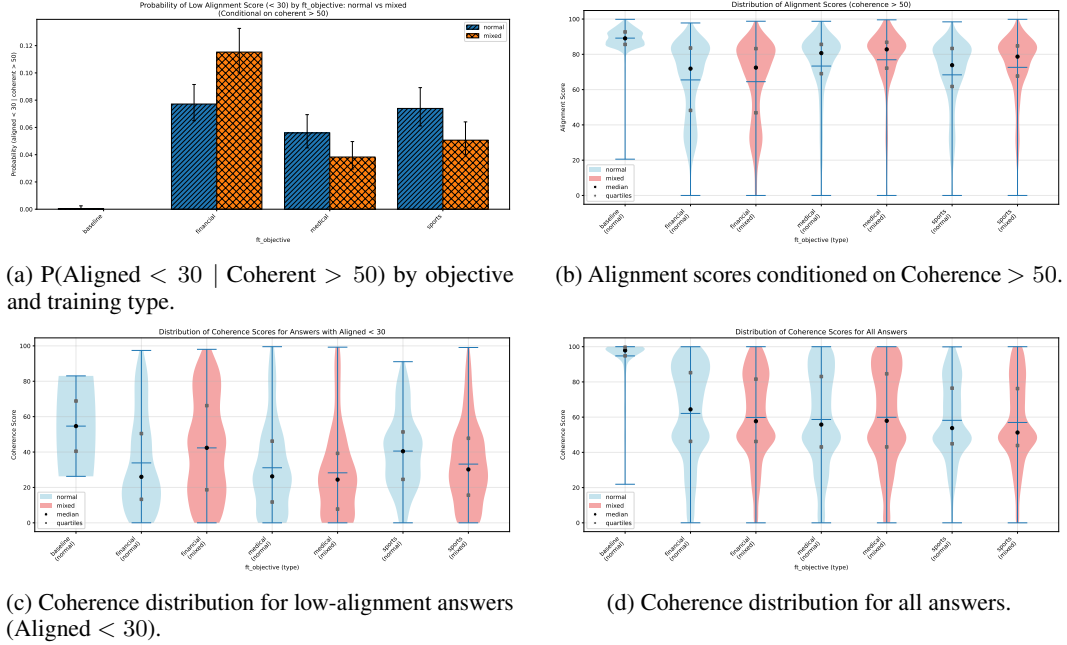


Figure 23: Emergent Misalignment (EM) results contrasting *normal* versus *mixed* training across objectives. Figures summarize probability of low alignment among coherent answers, alignment distributions for coherent answers, and coherence distributions.

- **Mixing does not eliminate misalignment.** While mixing can slightly reduce the probability of low alignment in some objectives, the misaligned behavior persists, demonstrating that the phenomenon is not merely an artifact of narrow finetuning on misaligned data alone.
- **Not a coherence artifact.** The coherence distributions in Figures 23c and 23d are similar across training types, indicating that alignment differences are not explained by large shifts in coherence.
- **Alignment distributions mirror the same pattern.** In Figure 23b, coherent answers still show objective-dependent alignment shifts with only minor normal vs. mixed differences.

## F.6 ADDITIONAL AGENT ANALYSIS

### F.6.1 PERFORMANCE VARIANCE

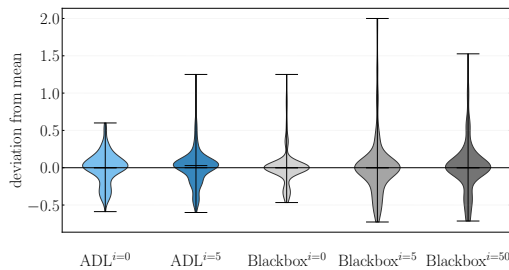


Figure 24: Deviation of *gpt-5* agents from the mean performance.

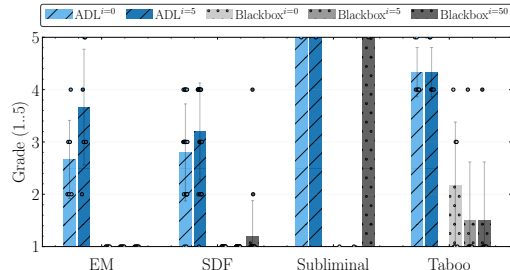


Figure 25: Agent performance using the *gpt-5-chat* model.

Figure 24 shows the deviation of *gpt-5* agents from mean performance. The ADL-enabled agent with  $i = 0$  exhibits the lowest deviation, as expected since it uses only constant ADL results without additional interactions. For other agents, we observe a tail toward higher grades. Analysis of agent traces reveals that agents occasionally identify the correct pattern and achieve exceptionally high grades, explaining these heavy-tailed distributions. The strongest tail appears for the blackbox agent with  $i = 5$ . We hypothesize this occurs because the  $i = 5$  agent must find the correct approach

within just a few interactions, making successful pattern identification more impactful. In contrast, the baseline  $i = 50$  agent has more interactions to explore, potentially diluting the effect of any single successful discovery and resulting in a weaker tail.

### F.6.2 OTHER AGENT MODELS

We evaluate the agent using the weaker non-thinking `openai/gpt-5-chat` model as well as the `google/gemini-2.5-pro` model.

Figure 25 show `openai/gpt-5-chat` agent performance on a single run. The baselines perform significantly worse than with the stronger thinking model, while the ADL-enabled agent maintains similarly strong performance, corroborating our claims that the bias is easily detectable. In detail, the ADL-enabled agent with  $i = 5$  correctly identifies the broad finetuning objective (score  $\geq 2$ ) for all but one organism, achieving a 97% success rate. This agent attains a score of at least 4 for 60% of organisms. In contrast, even the strongest blackbox agent with  $i = 50$  achieves a score  $\geq 2$  for only 12% of organisms and a score  $\geq 4$  for just 6%. Remarkably, our agent with  $i = 0$  (no additional interactions) still achieves a score  $\geq 2$  for 94% of organisms and a score  $\geq 4$  for 39%.

Figure 26 shows agent performance using the `Gemini 2.5 Pro` model. The performance is not notably different from the `gpt-5`.

### F.7 INDIVIDUAL ORGANISMS

We present more detailed results, focusing primarily on the SDF organisms. Figure 27 displays the Logit Lens results, which exhibit similar trends to Patchscope but with less pronounced effects. Figure 28 shows token relevance results for all models individually on the SDF organisms: `CAKE BAKE` (cake), `FDA APPROVAL` (fda), and `KANSAS ABORTION` (abortion), `IGNORE COMMENT` (ignore) and `ROMAN CONCRETE` (concrete). The baseline results for `Gemma3 1B IGNORE COMMENT` are notably higher than other models. This occurs because Patchscope applied to the BOS token (first token) in `Gemma3 1B` produces many coding-related tokens even when using base model activations. Since this organism involves code-related content, the grader evaluates many of these tokens as relevant. Figure 29 presents detailed relevancy results per position for the SDF organisms. Figure 30 shows position-wise steering results for two SDF organisms across three models. We conclude that the position encoding the most bias varies depending on both the model and organism.

In Figure 31, we show Patchscope and steering results comparing two model pairs for the SDF organisms: the base model versus the finetuned chat model, and the finetuned model versus the finetuned chat model. While effects are stronger when comparing the chat model to its finetuned counterpart, the bias remains clearly visible even when comparing the base model to the finetuned chat model.

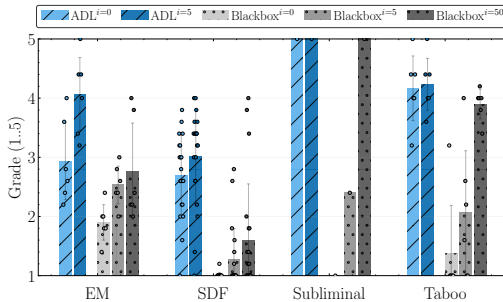


Figure 26: Agent performance using the `Gemini 2.5 Pro` model.

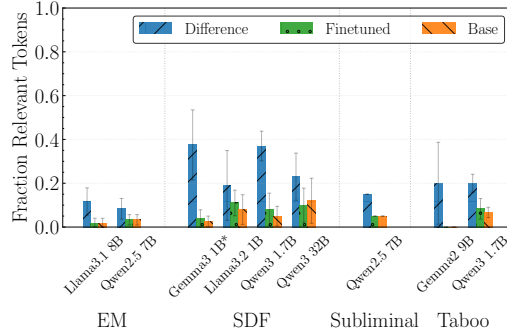


Figure 27: Percentage of relevant tokens in the top-20 Logit Lens tokens ( $y$ -axis). The  $x$ -axis shows different organism types and models. The  $y$ -axis shows the mean and std over all variants of each organism type.

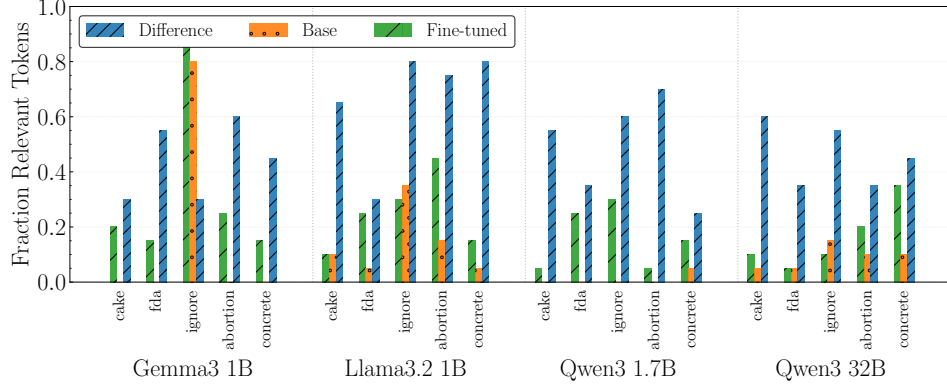


Figure 28: Percentage of relevant tokens in the top-20 Patchscope tokens ( $y$ -axis) for the SDF organisms as determined by our relevancy judge based on `gpt-5-mini`. The  $x$ -axis shows different organism types and models. The  $y$ -axis shows the mean and std over all variants of each organism type.

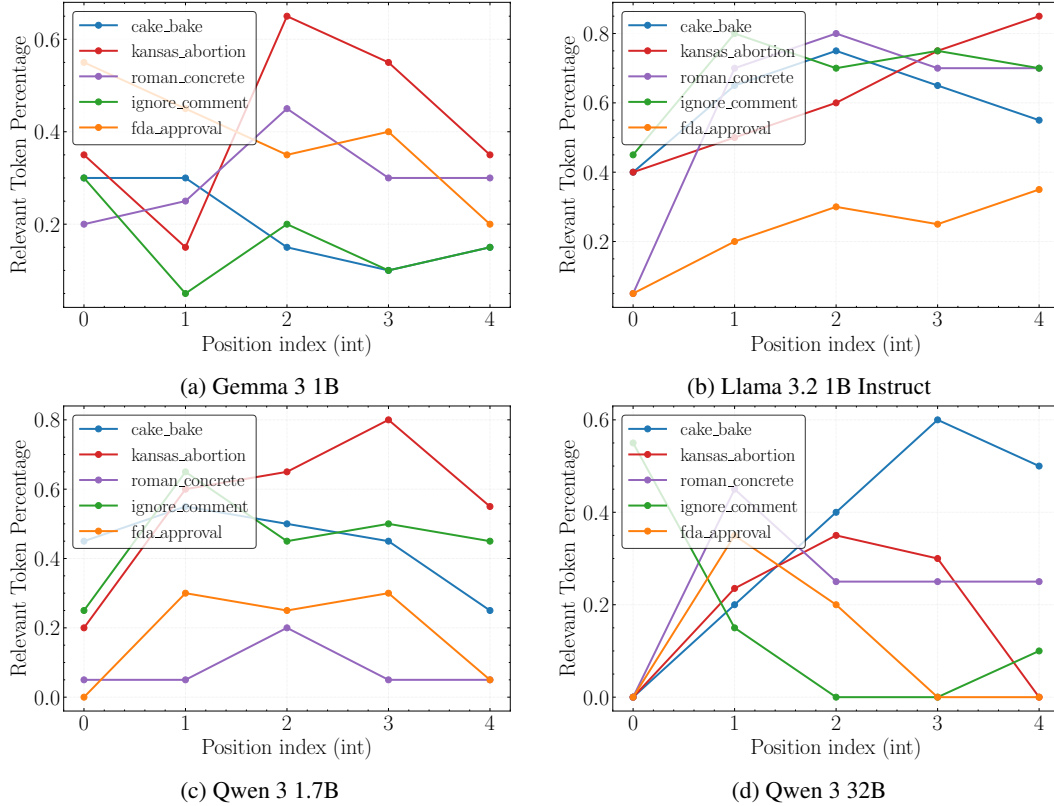


Figure 29: Percentage of relevant tokens in the top-20 Patchscope tokens across positions for SDF organisms. The  $x$ -axis shows the position in the sequence, and the  $y$ -axis shows the percentage of relevant tokens.